Basis of Fundamental Subspaces

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Finding the Column Space and its Basis

Example # 1:

Given the matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

We are asked to find the column space of A and its basis.

Step 1: Write the matrix in terms of its columns

The matrix A can be written as:

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

The column space of A is the span of the columns of the matrix. To find the basis of the column space, we need to check if these columns are linearly independent.

Step 2: Perform row reduction

We row reduce the matrix A to find its rank and determine which columns are linearly independent.

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 3 & 6 & 9 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The row-reduced echelon form of the matrix shows that the rank of A is 1, meaning there is only 1 linearly independent column.

Step 3: Identify the linearly independent columns

The first column is a pivot column, so it is linearly independent. The second and third columns are multiples of the first column, so they are dependent.

Step 4: Write the basis for the column space

Since the first column is linearly independent, the basis for the column space of A is given by:

$$\text{Basis for Col}(A) = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$$

Step 5: Conclusion

The column space of A is spanned by the vector $\begin{bmatrix}1\\2\\3\end{bmatrix}$ i.e., set of infinitely many elements, and the basis consists of this single vector.

Example # 2

Consider the matrix:

$$B = \begin{bmatrix}1 & 0 & 3\\2 & 1 & 6\\-1 & 1 & -3\end{bmatrix}$$

We are tasked with finding the column space of B and its basis.

Step 1: Write the matrix in terms of its columns

The matrix B can be written as:

$$B = \begin{bmatrix}1\\2\\-1\end{bmatrix} \quad \begin{bmatrix}0\\1\\1\end{bmatrix} \quad \begin{bmatrix}3\\6\\-3\end{bmatrix}$$

The column space of B is the span of the columns of the matrix. To find the basis, we need to check for linear independence.

Step 2: Perform row reduction

We apply Gaussian elimination to reduce the matrix and find the rank:

$$\begin{bmatrix}1 & 0 & 3\\2 & 1 & 6\\-1 & 1 & -3\end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix}1 & 0 & 3\\0 & 1 & 0\\-1 & 1 & -3\end{bmatrix} \xrightarrow{R_3 \to R_3 + R_1} \begin{bmatrix}1 & 0 & 3\\0 & 1 & 0\\0 & 1 & 0\end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix}1 & 0 & 3\\0 & 1 & 0\\0 & 0 & 0\end{bmatrix}$$

The row-reduced echelon form of the matrix shows that the rank of B is 2, indicating two linearly independent columns.

Step 3: Identify the linearly independent columns

From the row-reduced matrix, we observe that the first and second columns are pivot columns. Therefore, the first and second columns are linearly independent.

Step 4: Write the basis for the column space

Since the first and second columns are linearly independent, the basis for the column space of B is given by the vectors corresponding to the first and second columns of the original matrix B:

$$\text{Basis for Col}(B) = \left\{ \begin{bmatrix}1\\2\\-1\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix} \right\}$$

Step 5: Conclusion

The column space of B is spanned by the vectors $\begin{bmatrix}1\\2\\-1\end{bmatrix}$ and $\begin{bmatrix}0\\1\\1\end{bmatrix}$, and these two vectors form the basis for the column space.

Finding the Row Space and its Basis

Example

Consider the matrix:

$$B = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ -1 & -1 & -1 \end{bmatrix}$$

We will find the row space of B and its basis.

Step 1: Perform row reduction

We apply Gaussian elimination to reduce the matrix:

$$\begin{bmatrix} 2 & 4 & 6 \\ 1 & 3 & 5 \\ -1 & -1 & -1 \end{bmatrix} \xrightarrow{R_2 \to R_2 - \frac{1}{2}R_1} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 1 & 2 \\ -1 & -1 & -1 \end{bmatrix} \xrightarrow{R_3 \to R_3 + \frac{1}{2}R_1} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to \frac{1}{2}R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

The row-reduced echelon form shows two non-zero rows.

Step 2: Identify the linearly independent rows

The non-zero rows of the row-reduced matrix represent the linearly independent rows of the original matrix B.

Step 3: Write the basis for the row space

The basis for the row space of B is:

$$\text{Basis for Row}(B) = \left\{ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 \end{bmatrix} \right\}$$

OR

$$\text{Basis for Row}(B) = \left\{ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \begin{bmatrix} 1 & 3 & 5 \end{bmatrix} \right\}$$

Important Note:

Both basis are same as they are linear combination of each other but this is not the case for basis of column space as columns in echelon form are not linear combination of columns of A

Conclusion

The row space of B is spanned by the rows $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$.

Finding the Null Space

Example

Consider the matrix:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 3 & 6 & -3 \end{bmatrix}$$

We are tasked with finding the null space of A, which consists of all vectors $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{0}$.

Step 1: Set up the equation

The equation $A\mathbf{x} = \mathbf{0}$ can be written as:

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 3 & 6 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives us the following system of equations:

$$1x_1 + 2x_2 - 1x_3 = 0$$
$$2x_1 + 4x_2 - 2x_3 = 0$$
$$3x_1 + 6x_2 - 3x_3 = 0$$

Step 2: Perform row reduction

We reduce the augmented matrix to row echelon form:

$$\begin{bmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -2 & 0 \\ 3 & 6 & -3 & 0 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 3 & 6 & -3 & 0 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The row-reduced form gives us the equation:

$$x_1 + 2x_2 - x_3 = 0$$

Step 3: Express free variables

Let $x_3 = t$, a free variable. Then:

$$x_1 = -2x_2 + x_3 = -2x_2 + t$$

So the general solution to the system is:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_2 + t \\ x_2 \\ t \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Step 4: Write the basis for the null space

The basis for the null space of A is:

$$\text{Basis for Null}(A) = \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Conclusion

The null space of A is spanned by the vectors $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

Finding the Left Null Space

Example

Consider the matrix:

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 5 & 10 \end{bmatrix}$$

We are tasked with finding the left null space of B. The left null space consists of all vectors $\mathbf{y}$ such that $\mathbf{y}^T B = \mathbf{0}$.

Step 1: Set up the equation

The equation $\mathbf{y}^T B = \mathbf{0}$ can be written as:

$$\begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ 5 & 10 \end{bmatrix} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

This results in the system of equations:

$$y_1(1) + y_2(3) + y_3(5) = 0$$
$$y_1(2) + y_2(6) + y_3(10) = 0$$

Step 2: Write the system of equations

The system of equations is:

$$y_1 + 3y_2 + 5y_3 = 0$$
$$2y_1 + 6y_2 + 10y_3 = 0$$

Step 3: Perform row reduction

We form the augmented matrix and reduce it:

$$\begin{bmatrix} 1 & 3 & 5 & 0 \\ 2 & 6 & 10 & 0 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This gives us the equation:

$$y_1 + 3y_2 + 5y_3 = 0$$

Step 4: Express free variables

Let $y_3 = t$ (a free variable), and then express y_1 and y_2 in terms of y_3:

$$y_1 = -3y_2 - 5y_3$$

So the general solution is:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = t \begin{bmatrix} -5 \\ 0 \\ 1 \end{bmatrix} + y_2 \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$$

Step 5: Write the basis for the left null space

The basis for the left null space of B is:

$$\text{Basis for Left Null}(B) = \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Conclusion

The left null space of B is spanned by the vectors $\begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -5 \\ 0 \\ 1 \end{bmatrix}$.

Finding the Basis of a Subspace

Example # 1:

Let W be the subspace of $\mathbb{R}^4$ spanned by the following vectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

We are tasked with finding a basis for W.

Step 1: Write the matrix with the given vectors as rows

We form a matrix A where the rows are the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Step 2: Perform row reduction

We apply Gaussian elimination to row reduce the matrix A:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 1 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & -2 & -2 & -4 \end{bmatrix} \xrightarrow{R_3 \to -\frac{1}{2}R_3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow{R_1 \to R_1 - 2R_3} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

After row reduction, the matrix is in row echelon form.

Step 3: Identify the linearly independent rows

The non-zero rows of the row-reduced matrix represent a basis for the subspace W. Therefore, the basis for W is:

$$\text{Basis for } W = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix} \right\}$$

Conclusion

The basis of the subspace W is formed by the vectors $\begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$. Since these vectors are linearly independent and span the subspace W, they form a basis.

Example # 2: Finding the Basis of a Subspace in $\mathbb{R}^3$

Let W be the subspace of $\mathbb{R}^3$ spanned by the vectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$$

We are tasked with finding a basis for W.

Step 1: Write the matrix with the given vectors as rows

We form a matrix A where the rows are the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix}$$

Step 2: Perform row reduction

We apply Gaussian elimination to row reduce the matrix A:

$$\begin{bmatrix} 1 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This is now in row echelon form.

Step 3: Identify the linearly independent rows

The non-zero rows of the row-reduced matrix represent a basis for the subspace W. Therefore, the basis for W is:

$$\text{Basis for } W = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Conclusion

The basis of the subspace W is formed by the vectors $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

Example # 3: Finding the Basis of a Subspace in $\mathbb{R}^4$

Let W be the subspace of $\mathbb{R}^4$ spanned by the vectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 6 \end{bmatrix}$$

We are tasked with finding a basis for W.

Step 1: Write the matrix with the given vectors as rows

We form a matrix A where the rows are the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 4 & 6 \end{bmatrix}$$

Step 2: Perform row reduction

We apply Gaussian elimination to row reduce the matrix A:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 4 & 6 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{bmatrix} \xrightarrow{R_3 \to R_3 - R_2} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This is now in row echelon form.

Step 3: Identify the linearly independent rows

The non-zero rows of the row-reduced matrix represent a basis for the subspace W. Therefore, the basis for W is:

$$\text{Basis for } W = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix} \right\}$$

Conclusion

The basis of the subspace W is formed by the vectors $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$.

Finding the Coordinate Vector

Example #1:

Let $\mathbb{R}^3$ be the vector space with basis:

$$B = \left\{ \mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{b}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

and let $\mathbf{v}$ be the vector:

$$\mathbf{v} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

We want to find the coordinate vector of $\mathbf{v}$ with respect to the basis B.

Solution

To find the coordinate vector of $\mathbf{v}$ with respect to B, we express $\mathbf{v}$ as a linear combination of the basis vectors:

$$\mathbf{v} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + c_3\mathbf{b}_3$$

Substituting the basis vectors:

$$\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Equating components:

$$\begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Thus:

$$c_1 = 3, \quad c_2 = 4, \quad c_3 = 5$$

The coordinate vector of $\mathbf{v}$ with respect to the basis B is:

$$[\mathbf{v}]_B = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix}$$

Example# 2:

Let $\mathbb{R}^2$ be the vector space with basis:

$$B = \left\{ \mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

and let $\mathbf{v}$ be the vector:

$$\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

We want to find the coordinate vector of $\mathbf{v}$ with respect to the basis B.

Solution

To find the coordinate vector of $\mathbf{v}$ with respect to B, we express $\mathbf{v}$ as a linear combination of the basis vectors:

$$\mathbf{v} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2$$

Substituting the basis vectors:

$$\begin{bmatrix}3\\2\end{bmatrix} = c_1\begin{bmatrix}1\\1\end{bmatrix} + c_2\begin{bmatrix}1\\-1\end{bmatrix}$$

Equating components:

$$\begin{bmatrix}3\\2\end{bmatrix} = \begin{bmatrix}c_1 + c_2\\c_1 - c_2\end{bmatrix}$$

Solving the system of equations:

$$c_1 + c_2 = 3$$
$$c_1 - c_2 = 2$$

Adding the equations:

$$2c_1 = 5 \implies c_1 = \frac{5}{2}$$

Subtracting the equations:

$$2c_2 = 1 \implies c_2 = \frac{1}{2}$$

The coordinate vector of $\mathbf{v}$ with respect to the basis B is:

$$[\mathbf{v}]_B = \begin{bmatrix}\frac{5}{2}\\\frac{1}{2}\end{bmatrix}$$